



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

15.0 Why AI

July 2025

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Course Content

15. Why AI?	05
• Types of AI and its Branches	08
• AI vs Traditional Methods	11
Pre-Quiz	13

Learning Outcome

At the end of the session, you will be able to:

- Explain the basic fundamentals of data statistics and analytics.
- Describe the common terminologies of AI and its subsects, ML and DL.
- Apply data analytics, AI and their subsets usage in various ICM applications.
- Describe the characteristics of computing requirements most suitable for AI.

15.0 Why AI?

- Types of AI and its Branches
- Traditional Methods vs AI

Why AI? - Data Driven Decisions

Decision Making Paradigm

HUMAN CENTRIC DECISION MAKING



**Brain Computer
Interface**

Gesture Enabled AR / VR

Voice Enabled

Collaborative / Assisted

Self Service Analytics

MACHINE CENTRIC DECISION MAKING



**Artificial General
Intelligence**

Artificial Intelligence

Deep Learning

Machine Learning

Statistical Analysis

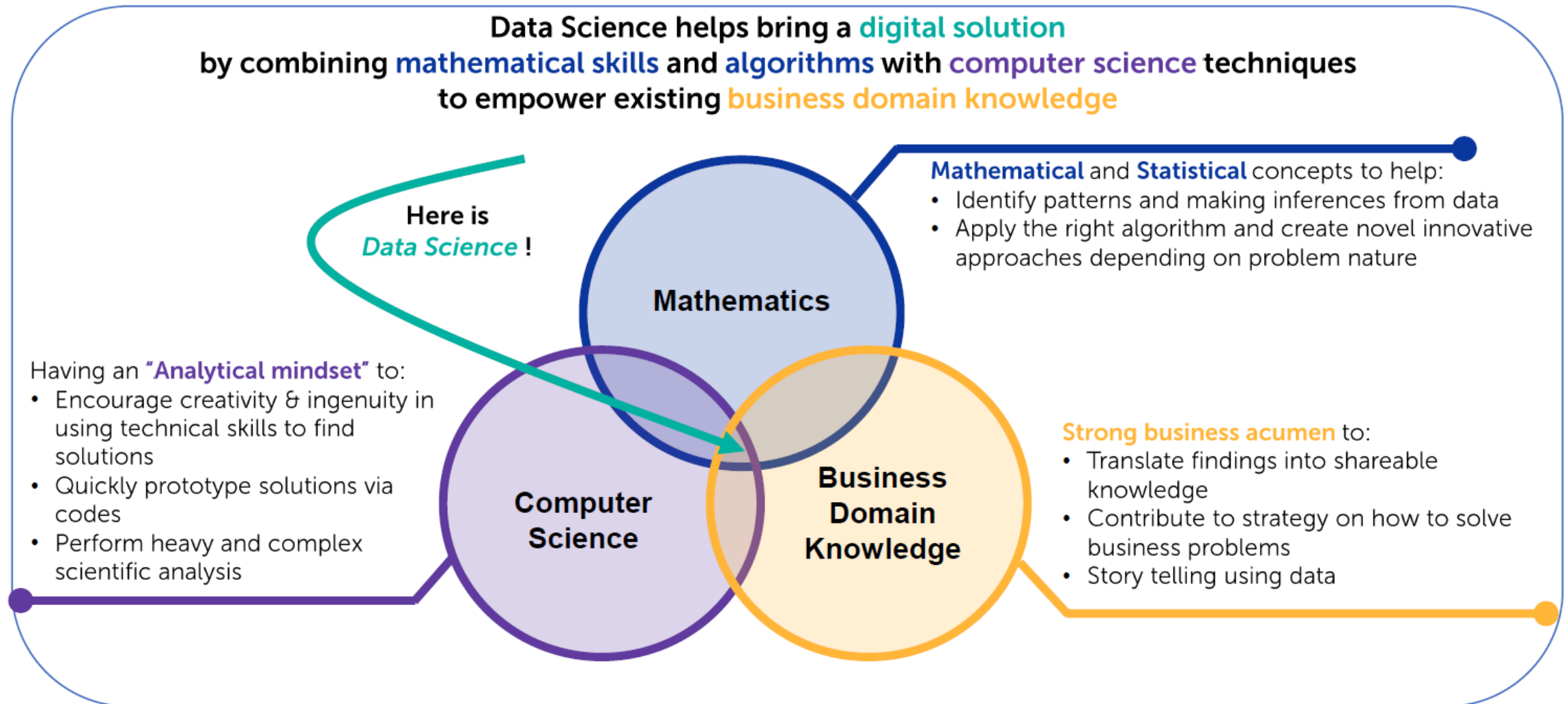
DATA FOUNDATION

Data: Structured, Semi-Structured, Unstructured (Text, Image, Video, Audio)
Information Management: DAM, Metadata, MDM, DQM, DS, ILM, DG

Source: PETRONAS Digital Academy

Why AI? - Data Driven Decisions

An organization should have analytics and pursue it for any data driven decisions



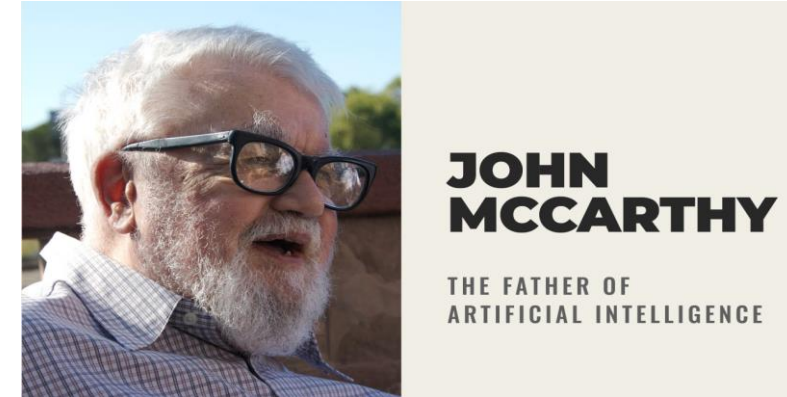
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Types of AI and its Branches

Artificial intelligence (AI) is the science and engineering of making intelligent machines, especially intelligent computer programs

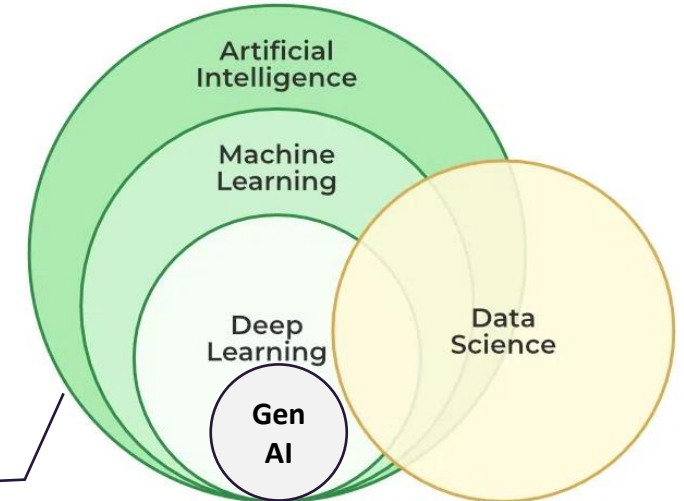
- Artificial Intelligence (AI) is the simulation of human intelligence by machines, particularly computer systems.
- It enables machines to perform tasks that typically require human cognition, such as reasoning, learning, problem-solving, perception, and language understanding.
- AI spans various fields and applications, revolutionizing industries and transforming daily life.



Data Science is the core of AI.
or
AI relies on Data Science.

3 Types of AI Based on Capabilities

- **Narrow AI (Weak AI):** Designed for specific tasks like voice assistants or recommendation systems.
- **General AI (Strong AI):** A theoretical form of AI capable of understanding and performing any intellectual task that a human can do.
- **Superintelligent AI:** A future concept where AI surpasses human intelligence across all fields.



Branches of AI

Source: <https://arxiv.org/pdf/2001.04561>

Branches of AI



Artificial intelligence (AI)

The simulation of human intelligence processes by machines, especially computer systems.



Machine learning (ML)

A subfield of AI focused on the use of data and algorithms in machines to imitate the way that humans learn, gradually improving its performance.



Deep learning (DL)

A machine learning technique that imitates the way humans gain certain types of knowledge; uses statistics and predictive modeling to process data and make decisions.



Generative AI

Algorithms (such as ChatGPT, DALL-E, Codex) that use prompts or existing data to create new content:

- Written: text, code
- Visual: images, videos
- Auditory: audio

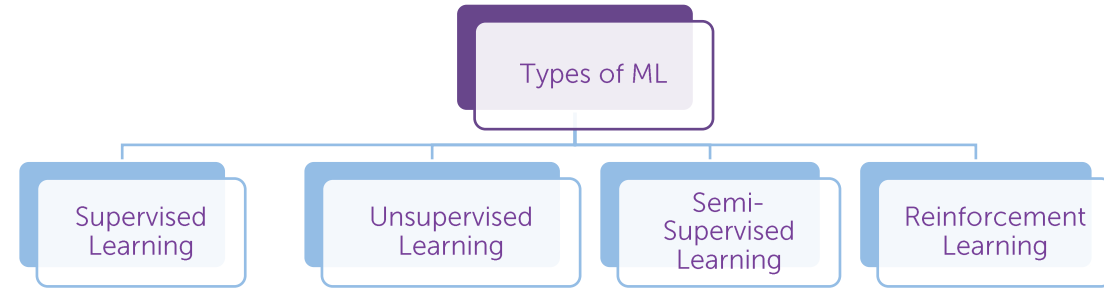
Used by data scientists

AI is now accessible to developers and end users

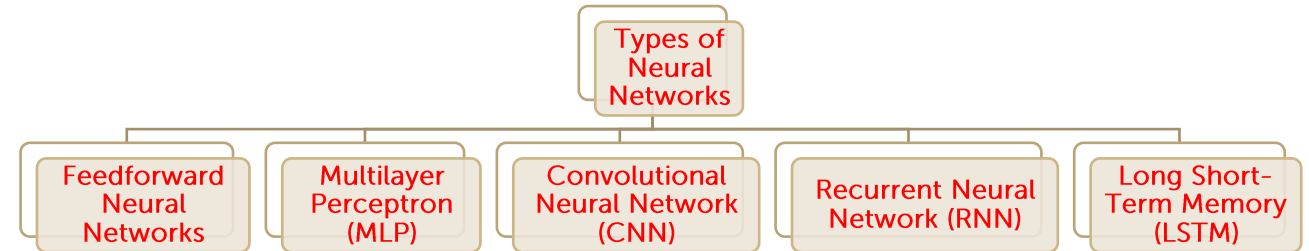
Branches of AI

Artificial Intelligence (AI) - AI is the science and engineering of making intelligent machines, especially intelligent computer programs.

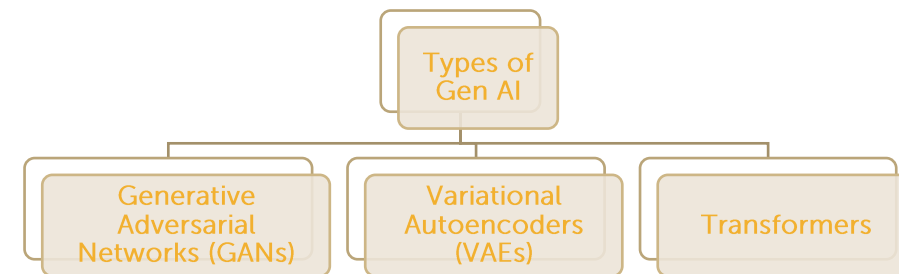
Machine Learning (ML) - ML is a subset of AI focused on developing systems that can learn and adapt from data without explicit programming.



Deep Learning (DL) - DL is a subset of ML that uses Artificial Neural Networks (ANN) with multiple layers to model and understand complex patterns in large datasets.



Generative AI (Gen AI) - Gen AI is a subset of DL that generates new content, such as text, images, music, code, and even videos, based on patterns learned from existing data.



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Traditional Methods vs AI

Traditional Methods	AI
Traditional Methods involves manual analysis using predefined rules and often uses smaller data.	AI used vast amount of data, uncovering complex patterns and making predictions.

Aspect	Traditional Methods	AI
Automation	Requires manual tuning and adjustments.	Learns and improves automatically with data.
Scalability	Limited by computational and analytical capacity.	Handles massive datasets with high dimensionality.
Pattern Recognition	Limited to linear or predefined relationships	Excels in identifying complex patterns.
Predictive Insights	Relies on predefined statistical models.	Highly accurate and data-driven forecasting.
Data Requirements	Works well with small, structured datasets.	Requires large datasets for training.
Explainability	Simple and easy to understand.	Can be challenging to interpret ("black box").
Real-time Processing	Limited in dynamic, real-time applications	Enables real-time decision-making.
Risk of Overfitting	Lower risk in controlled setups.	Higher risk without careful validation.
Key Applications	Descriptive analytics, process optimization.	NLP, image recognition, predictive maintenance.

Pre-Quiz

Interactive Learning

